Template to prepare preprints and manuscripts using markdown and github actions

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Purpose: This template provides a series of scripts to render a markdown document into an interactive website and a series of PDFs.

Internals: GitHub actions and a series of python scritpts. The markdown is handled with `pandoc`.

Motivation: It makes collaborating on text with GitHub easier, and means that we never need to think about the output.

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This template uses `pandoc` (and a few additional *Julia* glue scripts) to facilitate the production of scientific articles using a standard markdown file. The objective is to ensure that standard markdown (with the important exception of the `pandoc-crossref` citation markup) will be rendered into an interactive website (which allows collaborative annotations with the `hypothes.is` platform), a "draft" style PDF (double-spaced, numbered lines, figures at the end), and a "preprint" style PDF (with more reader-friendly pagination).

The core bit of configuration is the `metadata.json` file, which handles information about authorship, affiliations, the abstract, keywords, etc. All documents will be deployed to `gh-pages` *only* on push events from the `main` branch. All of the artifacts will be built when doing pull requests, so you can check that merging a branch is *not* going to cause the compilation of the documents to fail; indeed, you can download the artifacts produced during the run, to check the PDF and html files. The website is only updated from the `main` branch.

The workflow is *very* GitHub based, and so the manuscript file *is* the `README.md` - this is not going to be a huge issue as 90% of the markdown is standard, with the exception of the citations and mathematics, so this will render (mostly) like a normal README file.

Deploying the template

The process of deploying this template has been *greatly* streamlined from previous versions:

- Click on the "Use this template" button
- Edit `README.md` with your own text, commit, and push
- This push will trigger the first build - the builds are only active on the `main` branch (*not* `master`!), and on pull requests
- Go to `http://you.github.io/repo-name/` to view the html version, and get access to the PDFs
- Add your references to the `references.bib` file
- Edit the `metadata.json` file to add the title, abstract, authors

In particular, note that *you do not need* to create a personnal access token to deploy to `gh-pages` (from where the website is served).

The metadata file

General information

The title is a field in the `metadata.json`:

```
{
    "title": "Preprint template"
}
```

Authorship

Authors are listed as objects in the `authors` block. Each author is specified as follows:

```
{
      "familyname": "Bob",
      "givennames": "Alice",
      "email": "alice.bob@u.edu",
      "orcid": "0000-0000-0000-0001",
      "affiliations": [
        "Affiliation 1",
        "Affiliation 2"
      ],
      "status": ["corresponding", "equal"]
    }
```

The `email` field is recommended for all authors. The `status` field is only useful for the corresponding author, and to denote equal contributions. These informations are rendered on the initial page. If an `orcid` is given, it will be linked on the HTML and PDF versions.

Note that there is *no need* to number the affiliations - a small python script will take care of this automatically.

Abstract

This template supports three types of abstracts, indicated in the metadata file as `abstract`:

A regular `abstract` is defined as

```
"abstract": "A very long string"
```

An itemized abstract is an array of strings, each representing a bullet point:

```
"abstract": [
    "Point 1",
    "Point 2"
]
```

A structured abstract is an object with key-value pairs :

```
"abstract": {
    "Location": "Worldwide",
    "Organisms": "Mammals"
}
```

Citation style

The `citationstyle` key corresponds to the name, with `.csl` ommited, of a CSL stylesheet stored in the citation style language repository. Note that there is no difference between main and dependent styles, the build engine will take the correct steps to get the correct style. The default is `"citationstyle": "ecology-letters"`. There is a longer section about references management later on.

References management

The references are managed by `pandoc`. Note that we *do not* use `pandoc-citeproc`, which was an external module for older `pandoc` versions. References *must* be stored in a `references.bib` file, and that it would make sense to order it alphabetically by key.

We use Zotero for references management, and for the lab's manuscripts, we work from folders in a shared library (with a folder for every manuscript).

It is recommedned to use the Better BibTeX plugin for citation key generations, and auto-export of the shared library to the `references.bib` file. We use a citation key format meant to convey information on the author (first author full name), date (complet year), and title (first three letters of the first two non-stop words). It must be set in the Better BibTeX preferences as (you might need to remove the line changes):

```
auth.fold.fold + year + title.fold.nopunctordash.skipwords.lower.select(1,1).capitalize()
```

It is a good idea to configure Better BibTeX to auto-export on change, and to remove a lot of fields that are not strictly speaking required for references. The list of fields we usually ignore is:

```
abstract,copyright,annotation,file,pmid,month,shorttitle,keywords
```

The citations are done using the normal markdown syntax, where `@Elton1927AniEco` produces Elton (1927), and `[@Camerano1880EquViv]` produces (Camerano 1880).

Figures, Tables, and other floats

Note that you can wrap the text of legends for both figures and tables. This avoids the issue of having very long lines.

Mathematics

The following equation

$$J'(p) = \frac{1}{\log(S)} \times \left(-\sum p \times \log(p)\right) \tag{1}$$

is produced using

```
$$J'(p) = \frac{1}{\text{log}(S)}\times ... $$ {#eq:eq1}
```

and can be referenced using `@eq:eq1`, which will result in eq. 1. Note that because we use `pandoc-crossref`, the label "eq." will be generated automatically.

Tables

Table legends go on the line after the table itself. To generate a reference to the table, use `{#tbl:id}` – then, in the text, you can use `{@tbl:id}` to refer to the table. For example, the table below is tbl. 1. You can remove the *table* in front by using `!@tbl:id`, or force it to be capitalized with `\*tbl:id`.

Table 1: This is a table, and its identifier is `id` – we can refer to it using `{@tbl:id}`. Note that even if the table legend is written below the table itself, it will appear on top in the PDF document.

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa

Figures

Figures can have a legend – all figures *must* be in the `figures/` folder of the project, as it is also used for the website. We recommend to use good resolution images, rather than PDFs, or at least to have multiple versions available.

```
![This is the legend of the figure...](figures/figure.png){#fig:figure}
```

[Figure 1 about here.]

We can now use `@fig:figure` to refer to fig. 1.

Example text

Connectance, defined as the ratio of realized interactions on the total number of potential interactions, is one of the most common descriptor of network structure. In a bipartite network with T species at the top, and B at the bottom, having a total of L interactions, it is defined as $Co = L/(T \times B)$. Connectance has a lower bound, as the network cannot have fewer interactions that the number of species in its more speciose level – the minimal connectance is therefore $c_m = \max(T, B)$. This makes the connectance of networks of different sizes difficult to compare, especially since bipartite networks tends to have a low connectance. For this reason, we used a corrected version of connectance, defined as

$$Co^\star = \frac{L - c_m}{T \times B - c_m}\,. \tag{2}$$

This is a subsection

This takes values between 0 (the network has the minimal number of interactions) and 1 (all species are connected), but is robust to variations in species richness.

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Some non-standard maths

The phylogenetic reconstruction of $\hat{\mathscr{L}}$ and $\hat{\mathscr{R}}$ has an associated uncertainty, represented by the breadth of the uniform distribution associated to each of their entries. Therefore, we can use this information to assemble a *probabilistic* metaweb in the sense of (**Poisot2016StrPro?**), *i.e.* in which every interaction is represented as a single, independent, Bernoulli event of probability p.

Specifically, we have adopted the following approach. For every entry in $\hat{\mathscr{L}}$ and $\hat{\mathscr{R}}$, we draw a value from its distribution. This results in one instance of the possible left ($\hat{\ell}$) and right ($\hat{r}$) subspaces for the Canadian metaweb. These can be multiplied, to produce one matrix of real values. Because the entries in

$\hat{\ell}$ and $\hat{r}$ are in the same space where $\mathscr{L}$ and $\mathscr{R}$ were originally predicted, it follows that the threshold ρ estimated for the European metaweb also applies. We use this information to produce one random Canadian metaweb, $N = \hat{\mathscr{L}}\hat{\mathscr{R}}' \geq \rho$.

Because the intervals around some trait values can be broad (in fact, probably broader than what they would actually be), we repeat the above process 2×10^5 times, which results in a probabilistic metaweb P, where the probability of an interaction (here conveying our degree of trust that it exists given the inferred trait distributions) is given by the number of times where it appears across all random draws N, divided by the number of samples. An interaction with $P_{i,j} = 1$ means that these two species were predicted to interact in all 2×10^5 random draws, etc..

References

Camerano, L. (1880). Dell'equilibrio dei viventi merce la reciproca distruzione. *Atti Della R. Accad. Delle Sci. Torino*, 15, 393–414.

Elton, C.S. (1927). *Animal ecology*. University of Chicago Press.

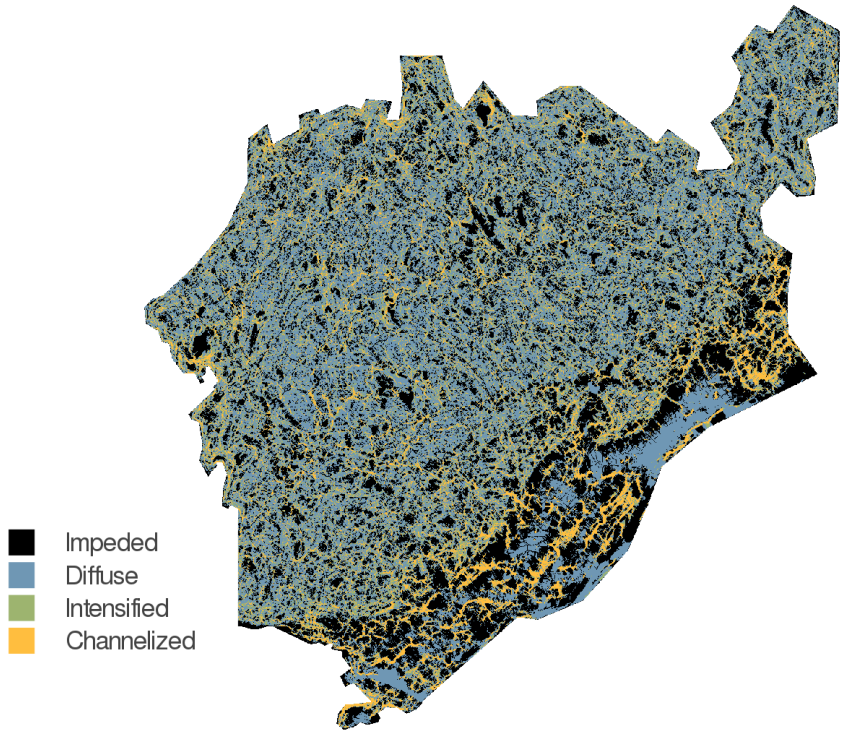

Figure 1: This is the legend of the figure, which will be shown in the margin in preprint mode, and underneath the figure in draft mode. The legend can contain references, etc. It is advised to use a resolution of at least 600dpi for the figures.